DeepShield

AI-Based Deepfake & Synthetic Face Detection Platform

Project ID: MCA21491
Complete Flowchart, Tech Stack aur Viva Explanation

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Language: Easy Hinglish (technical terms English me)

One-line project idea: User image ya video frame upload karta hai; system spatial face clues aur frequency-domain clues ko combine karke authenticity probability, manipulation region, generator category aur explainable heatmaps return karta hai.

1. Project ka Objective

DeepShield ka main purpose fake ya AI-generated faces ko authentic media se separate karna hai. Sirf ek black-box prediction dene ke bajay system batata hai ki suspicion kis facial region me hai, spatial evidence kya hai, frequency evidence kya hai aur final verdict kaise bana.

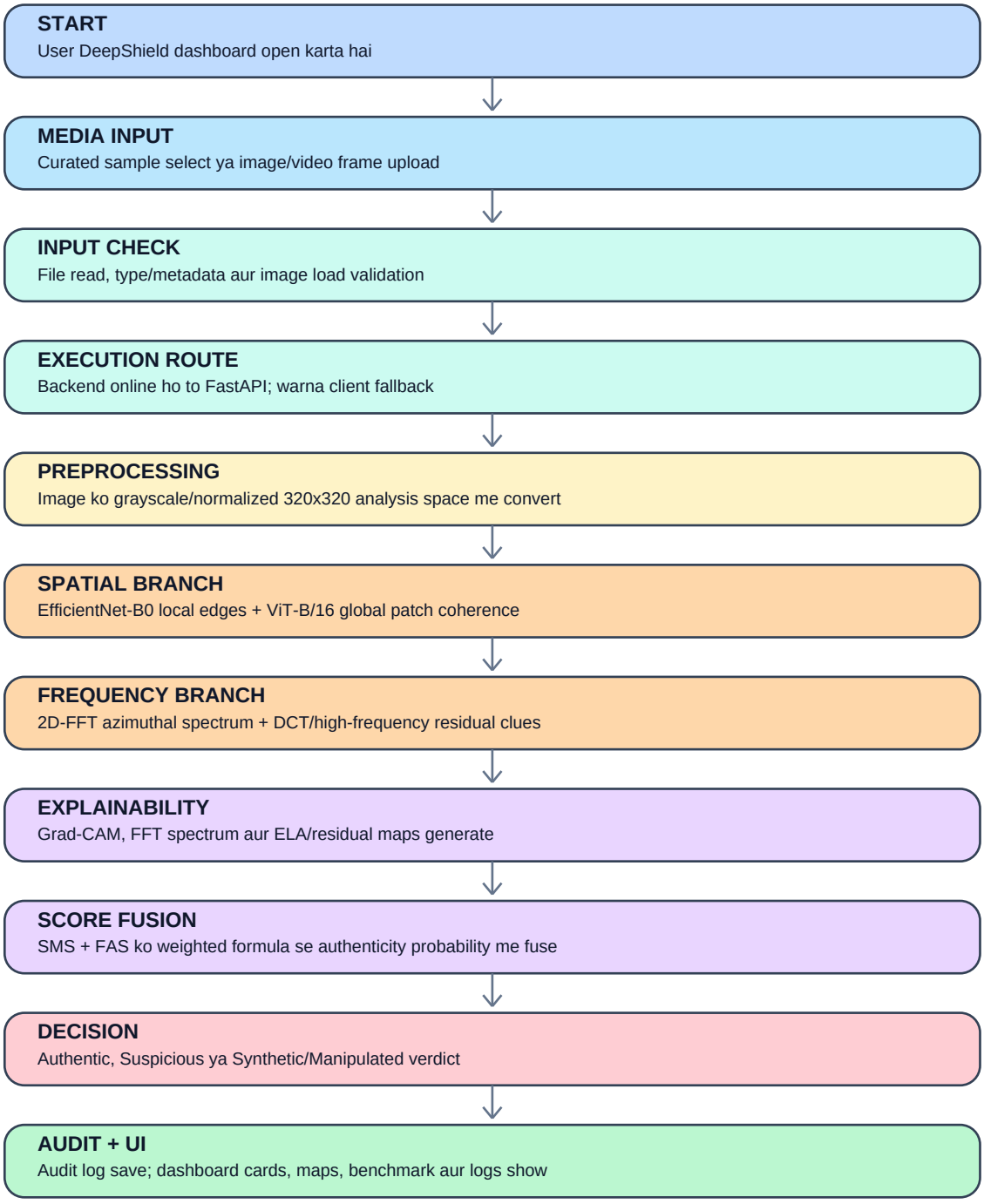
Problem kyun important hai?

GAN, Diffusion aur Face-Swap tools realistic faces bana sakte hain. KYC fraud, misinformation, identity misuse aur fake video ke case me human eye har baar reliable nahi hoti; isliye automated aur explainable forensic analysis useful hai.

Project ka output

Output me verdict (AUTHENTIC / SYNTHETIC / SUSPICIOUS), authenticity percentage, SMS, FAS, model confidence, primary manipulated region, generator category, Grad-CAM map, FFT spectrum, residual map, audit ID aur inference time milta hai.

2. End-to-End Master Flowchart



Flowchart ka short meaning: left-to-right nahi, yahan top-to-bottom pipeline hai; har box ek processing stage represent karta hai.

3. System ke Main Sections

Section	Kya hota hai?	Kyun hota hai / project me role
Forensic Workbench	Upload, sample selection aur analysis result screen.	Live investigation ke liye; user ek media ko inspect karta hai.
Benchmark	1,000 curated samples ke metrics aur model comparison.	System ki performance objectively dikhane ke liye.
Prisma Audit Logs	Past analysis ka ID, file, SMS, FAS, verdict aur time.	Traceability, repeat checking aur forensic record ke liye.
Architecture & Viva Guide	Dual-domain scoring, XAI aur future work ka explanation.	Sir ko design decisions aur viva answers samjhane ke liye.
Curated Samples	GAN, Diffusion, Face-Swap aur authentic DSLR examples.	Different manipulation types ko demo karne ke liye.

4. Detection Engine: Har Term ka Easy Explanation

Term	Easy Hinglish meaning	Is project me kya karta hai
EfficientNet-B0	Lightweight CNN jo local visual patterns dekhta hai.	Face boundary, sharp edge, blending aur texture artifacts ka spatial score S_eff nikalta hai.
ViT-B/16	Image ko 16x16 patches me todkar attention se global relation dekhta hai.	Eye, skin, mouth aur face patches ke beech natural coherence check karta hai; S_vit deta hai.
2D-FFT	Image ko spatial pixels se frequency pattern me convert karta hai.	GAN/Diffusion upsampling ke periodic high-frequency spikes aur spectral roll-off detect karta hai.
DCT / residual	Image ke high-frequency blocks aur reconstruction error ko inspect karna.	Checkerboard ya compression ke baad bache artificial patterns ko highlight karta hai.
Grad-CAM	Model evidence ko heatmap ke form me visualise karna.	Ocular, perioral aur jawline jaise suspicious facial regions highlight karta hai.
ELA / high-pass map	Image ke local error/edge difference ka visual map.	Manipulation boundary ya unnatural edge transition ko samjhata hai.
SMS	Spatial Manipulation Score, 0 se 1.	Local CNN aur global ViT evidence combine karta hai.
FAS	Frequency Anomaly Score, 0 se 1.	Frequency-domain anomaly ki strength batata hai.

5. Scoring Formula aur Decision Logic

System do independent domains ko combine karta hai, isliye sirf pixels ya sirf frequency par depend nahi karta. Isse explainability aur robustness improve hoti hai.

Spatial Manipulation Score: SMS = 0.55 x S_eff + 0.45 x S_vit

Yahan EfficientNet local boundary artifacts ko 55% weight deta hai aur ViT global patch coherence ko 45% weight deta hai.

Composite Authenticity Probability: A = (1 - (0.62 x SMS + 0.38 x FAS)) x 100

Manipulation risk badhega to authenticity probability kam hogi. Frontend me alphaSpatial slider isi balance ko explore karne deta hai; default spatial weight 0.62 hai.

Decision	Condition / meaning
SYNTHETIC / MANIPULATED	Manipulation probability 55% ya usse zyada; strong synthetic evidence.
SUSPICIOUS / REQUIRES MANUAL AUDIT	42% se 55% ke beech; automatic conclusion avoid karke human review suggest hota hai.
AUTHENTIC MEDIA	Manipulation probability 42% se kam; natural sensor/texture evidence stronger hai.

6. Backend Flow

Backend FastAPI par REST API provide karta hai. Startup par SQLite audit database initialize hota hai; upload aane par raw bytes read, forensic engine evaluate, audit ID create, record insert aur JSON result return hota hai.

Endpoint	Kya karta hai
GET /api/v1/health	Backend operational hai ya nahi aur active pipeline/version batata hai.
POST /api/v1/analyze	Image/video frame receive karke full forensic analysis aur audit ID return karta hai.
GET /api/v1/benchmark	1,000-sample benchmark summary return karta hai.
GET /api/v1/audit-logs	Recent 30 audit records dashboard ko deta hai.

Important: Demo mode me frontend client-side fallback engine bhi hai, isliye backend down hone par curated samples aur live upload analysis UI me chal sakte hain.

7. Frontend Flow aur UI Explanation

Frontend Next.js 14 + React + TypeScript par bana hai. page.tsx tab state maintain karta hai, sample/preset select karta hai, client maps generate karta hai aur result ko four tabs me present karta hai.

UI part	One-line explanation
Health indicator	FastAPI health endpoint call karke backend online/offline status dikhata hai.
Preset selector	Four known cases: StyleGAN3, Diffusion, DeepFaceLab video frame aur authentic DSLR.
Upload button	User ki local file read karta hai; backend available ho to POST bhi karta hai.
Quad view	Original, Grad-CAM, 2D-FFT aur ELA/residual ko ek saath compare karta hai.
Score cards	Authenticity, manipulation, SMS, FAS, EfficientNet, ViT aur confidence show karta hai.
Localization panel	Primary region aur 14x14 ViT + 7x7 EfficientNet map resolution explain karta hai.
Benchmark tab	Accuracy, precision, recall, F1, ROC-AUC aur comparison table show karta hai.
Audit tab	Past records ko compact forensic table me show karta hai.
Architecture tab	Viva ke liye design ke three core points aur future work dikhata hai.

8. Tech Stack

Layer	Technology	Use
Frontend	Next.js 14, React 18, TypeScript	Fast dashboard, component UI, type-safe state and browser execution.
Styling	Tailwind CSS, PostCSS	Clinical dark forensic theme, responsive layout and utility styling.
Icons	Lucide React	UI actions and status icons.
Client analysis	HTML5 Canvas, Sobel/Laplacian style maps, client FFT visualization	Backend unavailable ho to demo continuity and explainable maps.
Backend API	Python 3.11, FastAPI, Uvicorn	REST endpoints, upload handling and health service.
Forensics	NumPy, Pillow, OpenCV headless	Array/image processing and forensic feature generation.
Async option	Celery + Redis	Heavy analysis ko queue/worker me run karne ka scalable path.
Database	SQLite local; PostgreSQL production; Prisma schema	Audit logs and benchmark evaluation records.
Deployment	Docker Compose	Frontend, backend, worker, Redis and PostgreSQL services ko coordinate karta hai.

9. Benchmark aur Performance

Benchmark generator ke according total 1,000 samples hain: 500 authentic aur 500 synthetic. Synthetic set me GAN, Diffusion aur Face-Swap categories include hain; different JPEG, WebP aur H.264 compression conditions bhi include ki gayi hain.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
EfficientNet-B0	94.20%	93.85%	94.60%	94.22%	0.9810
ViT-B/16	95.10%	94.92%	95.30%	95.11%	0.9865
DeepShield Hybrid	98.70%	98.52%	98.89%	98.71%	0.9990

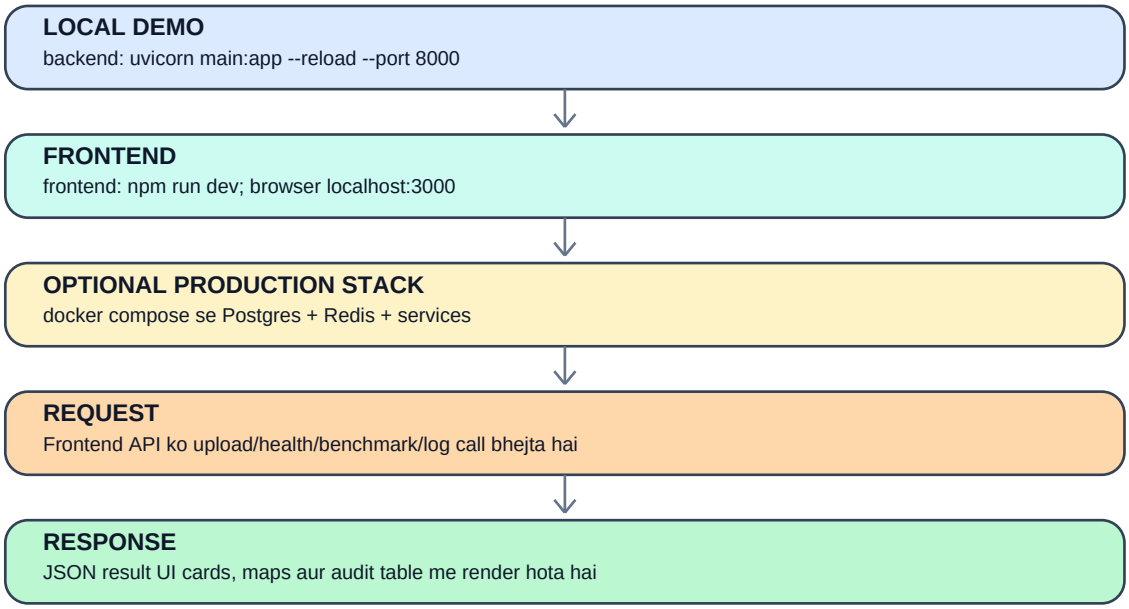
Metrics ka meaning: Accuracy = total correct; Precision = flagged samples me kitne actually fake; Recall = actual fake me kitne pakde; F1 = precision aur recall ka balance; ROC-AUC = ranking/separation quality.

10. Database aur Audit Trail

Local quick demo SQLite file me media_audit_logs table use karta hai. Production design Prisma ke MediaAuditLog aur BenchmarkEvaluation models ko PostgreSQL datasource ke saath define karta hai. Har analysis ko unique auditCode, filename, media type, region, compression, category, SMS/FAS, authenticity, verdict, latency aur timestamp ke saath store kiya jata hai.

Audit trail ka fayda: result later reproduce/check ho sakta hai, sir ko demo history dikh sakti hai aur forensic workflow me accountability maintain hoti hai.

11. Deployment / Run Flow



12. Limitations aur Future Work

Current project explainable demo/prototype architecture hai. Next phase me MesoNet + FFT/DCT fusion, WhatsApp/Instagram extreme recompression robustness, MediaPipe multi-frame eye-blink temporal verification, optical flow consistency aur real-time video stream support add kiya ja sakta hai.

13. Sir ko Explain Karne ke Liye Ready Viva Lines

Question	Answer line
Project kya karta hai?	DeepShield image/video frame ke spatial aur frequency evidence ko combine karke explainable deepfake detection karta hai.
Do domains kyun?	Spatial branch visible/local face artifacts dekhti hai, frequency branch generator ke hidden upsampling/compression patterns; dono milkar false decisions reduce karte hain.
EfficientNet aur ViT saath kyun?	EfficientNet local edges/textures me strong hai, ViT long-range patch relationships me; roles complementary hain.
Grad-CAM kyun?	Prediction ke saath evidence location dikhane ke liye, taaki output black-box na rahe.
FFT kyun?	AI-generated images me periodic spectral artifacts pixels se kam obvious hote hain, frequency view unhe expose karta hai.
Backend down ho to?	Frontend ka client-side Canvas fallback demo ko continue karta hai; backend online ho to FastAPI result aur audit storage use hota hai.
Audit log ka purpose?	Har analysis ka traceable record maintain karna, later review aur accountability ke liye.
Benchmark ka purpose?	Model ko bas claim se nahi, fixed sample set aur standard metrics se evaluate karna.
Main contribution?	Hybrid dual-domain score + explainable heatmaps + practical fallback dashboard + audit trail ko ek platform me integrate karna.

Final summary: DeepShield ka complete flow hai: Input -> Preprocess -> Spatial analysis -> Frequency analysis -> Explainability maps -> Score fusion -> Verdict -> Audit log -> Dashboard presentation. Isliye project detection ke saath reasoning bhi provide karta hai.